Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Immigration compliance management for foreign tourists visiting Sri Lanka remains a largely manual and fragmented process. Travelers from diverse nationalities must independently navigate complex Electronic Travel Authorization (ETA) requirements, visa regulations, and document preparation, entirely disconnected from their actual travel plans. This disconnect frequently results in invalid travel arrangements, overstay risks, and avoidable entry denials, undermining Sri Lanka's tourism growth which recorded only 2.5 million arrivals against a 3 million target in 2025, generating US$4 billion in revenue below its full potential.

Existing tools such as government immigration portals and third-party visa helper platforms provide only static, generic information. They cannot reason over a tourist's specific nationality, travel duration, and visit purpose in real time, and remain completely isolated from AI-driven travel planning ecosystems. This represents a critical research gap that blocks the realization of truly intelligent and legally sound tourism experiences.

This research proposes an AI-Based Immigration and Travel Compliance Management System a rule-based AI decision engine integrated directly into the broader Smart Tourism Management System. The module automatically extracts tourist nationality and travel details from user profiles, evaluates visa and ETA eligibility in real time, and generates a personalized compliance checklist with required documents and timelines. It prevents legally infeasible itineraries from being confirmed and delivers automated alerts for visa expiry risks and missing documentation.

The system utilizes Rule-Based AI logic, Natural Language Processing for profile extraction, a centralized regulatory knowledge base derived from Sri Lanka's Department of Immigration and Emigration guidelines, and REST API integration for live compliance validation.
Expected outcomes include a measurable reduction in tourist immigration errors and seamless embedding of legal compliance within the travel planning journey. Socially, it enhances visitor confidence and supports national tourism governance. Commercially, it reduces legal risk for tour operators and contributes to Sri Lanka's broader digital tourism transformation.

Keywords: Immigration Compliance, Rule-Based AI, ETA Verification, Smart Tourism, Travel Decision Support.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ADSS	Adaptive Decision Support System
AI	Artificial Intelligence
API	Application Programming Interface
ETA	Electronic Travel Authorization
ML	Machine Learning
REST	Representational State Transfer
NLP	Natural Language Processing
SST	Software Systems and Technologies
UI	User Interface
UX	User Experience

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1. INTRODUCTION

Immigration compliance is recognized as one of the most persistent barriers in international tourism, with visa complexity and entry regulation failures affecting millions of travelers annually across global destinations [1] [2]. For a developing economy like Sri Lanka, where tourism contributes 4-5% of GDP and over 300,000 jobs depend on the sector, the consequences of compliance failures extend beyond individual inconvenience to measurable national economic loss [3]. Foreign tourists visiting Sri Lanka must navigate a fragmented landscape of Electronic Travel Authorization (ETA) requirements, visa categories, and documentation obligations that vary significantly by nationality, duration, and purpose of visit a complexity that existing digital tools are entirely ill-equipped to manage [1]. Over the past decade, researchers have demonstrated that rule-based AI and NLP-driven decision systems can reliably automate complex legal reasoning and eligibility evaluation in regulated domains, yet these technologies remain completely unapplied to the tourism immigration compliance space [4] [5]. Current solutions such as government immigration portals and third-party visa platforms provide only static, disconnected guidance; they cannot reason over a tourist's specific travel profile in real time and offer no integration with travel planning or itinerary confirmation workflows [3]. This research proposes an AI-Based Immigration and Travel Compliance Management System designed specifically for Sri Lanka's regulatory context. The system integrates a rule-based AI decision engine, NLP-based profile extraction, a structured regulatory knowledge base, and an automated alert mechanism to deliver proactive, personalized compliance validation embedded directly within the travel planning process. Unlike existing tools, the proposed system prevents legally infeasible itineraries from being confirmed, generates nationality-specific compliance checklists, and proactively notifies tourists of visa deadlines addressing the real-time, integrated, and dynamic compliance gap that no current solution resolves.

1.1 Background And Context

Immigration compliance management is a persistent and structurally unresolved challenge in international tourism, directly affecting millions of foreign travelers who must independently navigate complex visa regulations, entry authorization requirements, and documentation obligations across global destinations [2]. This is of particular concern in Sri Lanka, where the tourism sector contributes 4–5% of national GDP, supports over 300,000 direct and indirect employment positions, and recorded 2.05 million tourist arrivals in 2023 and 2.4 million in 2024, both falling short of annual government targets due to persistent structural barriers including entry regulation complexity [3]. Foreign tourists visiting Sri Lanka are legally required to obtain an Electronic Travel Authorization (ETA) or an appropriate visa category prior to or upon arrival, with requirements varying significantly by nationality, duration of stay, and purpose of visit across more than 100 eligible nationalities a regulatory landscape that is complex, frequently updated, and poorly communicated through existing digital channels [1]. The consequences of compliance failures are significant across all stakeholders: tourists face detention at ports of entry, financial penalties, forced deportation, and complete disruption of pre-paid itineraries; tour operators are exposed to legal liability and reputational damage; and the Sri Lankan government bears increased administrative burden at border management institutions while its ambition to project Sri Lanka as a modern, well-regulated tourism destination is undermined [3] [2]. Existing digital solutions are fundamentally inadequate the official Department of Immigration and Emigration website provides only static, impersonalized information with no connection to travel planning workflows, while third-party visa platforms such as iVisa and VisaHQ function purely as information intermediaries that cannot prevent tourists from confirming legally invalid itineraries, and generic travel planning tools provide no immigration compliance functionality whatsoever [1]. Over the past decade, researchers have demonstrated that rule-based AI systems can reliably formalize and evaluate complex legal eligibility logic, and that NLP-driven profile extraction can accurately identify compliance-relevant attributes from free-text inputs, yet these technologies remain completely unapplied to the tourism immigration compliance domain [4] [5]. This research proposes an AI-Based Immigration and Travel Compliance Management System specifically designed for Sri Lanka's regulatory context, integrating a rule-based AI decision engine, NLP-based tourist profile extractor, a structured regulatory knowledge base encoded from official immigration guidelines, a personalized compliance checklist generator, and an automated alert mechanism delivering proactive, real-time compliance validation embedded directly within the travel planning process to ensure that only legally sound itineraries are confirmed [6] [4] [7].

1.2 Literature Overview

The SLTDA Annual Statistical Report 2023 and Year in Review 2024 document the persistent gap between Sri Lanka's tourism arrival targets and actual outcomes, recording 2.05 million arrivals in 2023 and 2.4 million in 2024 against higher government targets, while identifying the absence of intelligent digital compliance tools as a contributing structural barrier [3]. These reports establish the empirical foundation for this research by confirming that immigration compliance failures are not isolated incidents but a nationally significant and economically consequential systemic problem. The WTTC Sri Lanka Economic Impact Report 2023 further confirms that Sri Lanka's travel and tourism GDP contribution reached USD 4.0 billion, with sustained growth projections remaining contingent on removing regulatory access barriers for international visitors [2]. Together these reports directly justify the urgent need for the system proposed in this research.

Regarding existing products, iVisa and VisaHQ represent the most widely used commercial visa

assistance platforms globally, offering visa application processing and documentary guidance for travelers visiting Sri Lanka and other destinations. However, both platforms function purely as reactive, transactional intermediaries they assist tourists in submitting visa applications after travel decisions have already been made but provide no integration with itinerary planning or booking confirmation systems [1]. Neither platform prevents tourists from confirming legally infeasible travel plans, nor do they generate real-time compliance warnings during the trip planning phase, meaning a tourist can fully book accommodation, transport, and activities without ever receiving a warning that their planned stay duration exceeds their authorized visa period.

The official Sri Lanka Department of Immigration and Emigration portal represents the primary government-maintained compliance resource for foreign tourists, providing nationality-based ETA eligibility information and documentation requirements [1]. While authoritative in content, the portal delivers only static, impersonalized information with no real-time validation capability and no connection to any travel planning ecosystem. The portal assumes tourists will proactively seek compliance information before booking an assumption that consistently fails in practice and provides no mechanism for generating personalized, action-oriented guidance tailored to an individual tourist's specific profile, travel dates, or purpose of visit.

Palmirani et al. (2011) demonstrated the effectiveness of formally encoding complex regulatory rules into machine-readable ontological formats for computational reasoning through the LegalRuleML framework [4]. Their study proved that immigration-style conditional eligibility rules structured as if-then logical conditions based on nationality, duration, and purpose can be represented as formal rule sets and evaluated through automated inference engines with high accuracy. This work provides the direct methodological foundation for the Regulatory Knowledge Base and Rule-Based AI Decision Engine components of the proposed system, establishing that Sri Lanka's visa and ETA regulations can be computationally encoded and queried in real time without requiring human adjudication for each compliance evaluation.

Vadisetty, Polamarasetti, and Kunchala (2026) demonstrated that automated AI systems can effectively monitor and enforce compliance obligations across complex multi-rule regulated environments, reducing human error rates and improving institutional adherence to legal requirements at scale [5]. Their study validated that AI-driven compliance monitoring is not only technically feasible but measurably more effective than manual or reactive approaches across regulated industries. This directly supports the core design premise of the proposed system that real-time, automated compliance validation embedded within the travel planning process will outperform any manual or static guidance approach currently available to tourists planning visits to Sri Lanka.

Vaswani et al. (2017) introduced the Transformer architecture, which forms the technical backbone of modern NLP systems capable of extracting structured information from unstructured free-text inputs with high accuracy [7]. Their model demonstrated that attention-based deep learning architectures can identify and classify named entities including locations, dates, durations, and intent categories from natural language text far more effectively than prior rule-based or recurrent neural network approaches. Applied to tourist profile inputs, this capability enables the NLP-Based Profile Extractor component of the proposed system to automatically identify nationality, travel dates, duration, and purpose of visit from natural language descriptions, eliminating the need for lengthy structured forms and making compliance validation seamless within the planning workflow.

Davis (1989) established through the Technology Acceptance Model that digital systems achieve

meaningful user adoption only when users simultaneously perceive them as both useful and easy to use [6]. Their study found that perceived ease of use is particularly critical in voluntary-use contexts where users have no obligation to engage with a system precisely the condition that applies to tourist-facing compliance tools. Applied to the compliance checklist generator of the proposed system, this principle informs the design decision to present all guidance as plain-language, action-oriented outputs embedded within the familiar itinerary confirmation interface rather than as a separate regulatory portal that tourists must independently navigate.

Though such developments can be identified, even the most advanced existing compliance tools, visa platforms, and AI regulatory systems share critical limitations when considered against the specific requirements of tourism immigration compliance management. Most existing systems are entirely reactive providing information only when tourists explicitly request it rather than proactively monitoring planning activity and intervening to prevent violations before itineraries are confirmed [1] [5]. Current commercial visa platforms such as iVisa and VisaHQ are completely disconnected from travel booking ecosystems, meaning legally infeasible trips can be fully confirmed and paid for without any compliance warning being generated. Existing tools operate at a generic nationality level and cannot produce personalized, document-specific compliance checklists tailored to an individual's exact travel profile. Furthermore, no existing solution combines rule-based regulatory reasoning, NLP-based profile extraction, real-time itinerary validation, and proactive automated alerts within a single integrated module embedded inside a broader smart tourism management system [4].

Under such limitations, it is evident that a proactively embedded, AI-powered immigration compliance system that goes beyond static information delivery is urgently needed for Sri Lanka's tourism ecosystem. The AI-Based Immigration and Travel Compliance Management System proposed in this research is designed to address all identified deficiencies through the integration of a structured regulatory knowledge base, a rule-based AI decision engine, an NLP-based tourist profile extractor, a personalized compliance checklist generator, and an automated alert and reminder engine delivering a proactive, personalized, and continuously embedded compliance experience that no current commercial or governmental solution provides.

1.3 Research Gap

The review of existing literature, official reports, and commercial solutions reveals a clear, significant, and evidence-supported research gap at the intersection of immigration compliance management and intelligent travel planning systems. Despite substantial progress in AI-driven tourism platforms, rule-based regulatory reasoning, and NLP-based information extraction, no existing system integrates immigration compliance validation proactively and directly within the travel itinerary planning process either for Sri Lanka or for comparable developing tourism economies [3] [8].

The first critical gap is the absence of proactive compliance intervention. All currently available solutions including the Sri Lanka Department of Immigration and Emigration portal, iVisa, and VisaHQ are entirely reactive, providing immigration information only when tourists explicitly request it [1]. Research has established that reactive compliance tools consistently fail to prevent violations because tourists do not seek compliance guidance proactively before confirming bookings [6] [9]. Nalbandian (2022) demonstrated through analysis of Canada's automated visa decision system that even government-operated AI immigration tools focus exclusively on backend processing efficiency, with no capability to deliver proactive, tourist-facing compliance guidance during the pre-booking planning phase [9]. This reactive-only approach represents a fundamental architectural failure that no current solution has addressed.

The second critical gap is the complete isolation of compliance tools from travel planning and itinerary confirmation workflows. Mustafa (2019) identified that embedding compliance guidance directly

within travel planning interfaces is the most impactful intervention available to reduce tourist regulatory failures in Sri Lanka, yet no such integration currently exists [10]. Bellodi, Zese, and Bertasi (2022) demonstrated that machine learning policy support systems for smart tourism management can effectively integrate regulatory data with operational tourism workflows, but their work focused exclusively on tourist flow optimization with no application to immigration compliance validation [8]. The gap between what is technically achievable as demonstrated by Bellodi et al. and what is actually deployed in the immigration compliance domain is precisely the research opportunity this study addresses.

The third critical gap is the absence of nationality-specific, personalized compliance intelligence. Existing tools operate at a generic level, providing the same information to all tourists regardless of their specific nationality, visa category, travel duration, or purpose of visit [1] [4]. The regulatory knowledge base framework established by Palmirani et al. (2011) demonstrates that multi-condition eligibility rules varying across individual profiles can be formally encoded and evaluated computationally with high accuracy [8], yet this capability has never been applied to build a personalized, tourist-facing compliance guidance system. No existing solution generates a dynamic, action-oriented compliance checklist tailored to an individual tourist's exact profile a gap directly confirmed by the absence of any such feature across all reviewed commercial and governmental platforms.

The fourth critical gap is the non-existence of automated proactive alert mechanisms for compliance deadline management within tourism platforms. Vadisetty et al. (2026) confirmed that automated AI compliance monitoring systems significantly reduce deadline violations in regulated industries [5], yet no tourism platform currently incorporates automated alerts for visa expiry risks, missing documentation, or compliance deadline tracking [3]. This means tourists who successfully obtain an ETA or visa receive no further intelligent guidance to maintain compliance throughout their stay, leaving the most time-sensitive phase of immigration management entirely unmanaged by any digital tool.

These four gaps collectively reveal that the research problem is not merely a missing feature in existing platforms but a system-level failure in Sri Lanka's digital tourism ecosystem. The proposed AI-Based Immigration and Travel Compliance Management System addresses all four identified gaps simultaneously delivering proactive compliance intervention, direct integration with itinerary confirmation, nationality-specific personalized guidance, and automated deadline alert delivery within a single embedded intelligent module. No prior research has proposed or evaluated a system that combines rule-based AI reasoning [4], NLP-based tourist profile extraction [7], itinerary-level compliance validation, and proactive alert delivery [5] within a tourism management platform, establishing this research as a novel, first-of-its-kind contribution to both the smart tourism and AI compliance management domains.

2. OBJECTIVE

2.1 Main Objective

To design and develop an AI-based mobile application for immigration and travel compliance management that automatically evaluates visa and ETA eligibility based on tourist nationality, travel duration, and purpose of visit, generates personalized compliance checklists, and delivers proactive automated alerts ensuring that only legally feasible itineraries are confirmed for foreign tourists visiting Sri Lanka.

2.2 Specific Objectives

1. To analyze and formalize Sri Lanka's visa and ETA regulations into a structured, machine-readable regulatory knowledge base covering nationality-specific eligibility conditions, documentation requirements, and authorized stay durations for a minimum of 10 major tourist source nationalities.
2. To design and implement a rule-based AI decision engine that evaluates tourist visa and ETA eligibility in real time by reasoning over nationality, travel duration, and purpose of visit, achieving a compliance verdict accuracy of at least 90% against manually verified test cases.
3. To develop an NLP-based tourist profile extractor that automatically detects nationality, travel dates, duration, and purpose of visit from free-text user inputs and itinerary data, reducing manual data entry requirements for compliance validation to zero.
4. To build a personalized compliance checklist generator that produces nationality-specific, action-oriented lists of required documents, applicable fees, and submission timelines, and to integrate compliance validation directly with the travel planning engine to prevent legally infeasible itineraries from being confirmed.
5. To implement an automated alert and reminder engine that delivers proactive mobile push notifications for visa expiry risks, missing documentation, and compliance deadlines, targeting a minimum 80% reduction in tourist-reported compliance deadline violations during usability testing.
6. To evaluate the overall system through functional testing, user acceptance testing, and expert review, measuring compliance accuracy, alert effectiveness, checklist usability, and itinerary validation reliability against defined performance benchmarks.

3. METHODOLOGY

3.1 Research Approach

This research adopts the **Design Science Research (DSR)** methodology, which is the most appropriate approach for studies that aim to design, develop, and evaluate a novel IT artefact as a solution to a real-world problem. DSR is widely applied in information systems and software engineering research where the primary contribution is a working system rather than a theoretical model alone. Following the DSR framework proposed by Hevner et al., this research proceeds through three iterative cycles: the relevance cycle (identifying the real-world immigration compliance problem), the design cycle (building and refining the AI-based compliance system), and the rigor cycle (grounding the design in established knowledge from rule-based AI, NLP, and compliance systems literature). The research is applied and experimental in nature the system is built, tested against defined performance benchmarks, and evaluated through functional testing and user acceptance testing to validate that it solves the stated problem effectively.

3.2 Technical Methods and Algorithms

The proposed system is built on four core technical methods:

Rule-Based AI Inference Engine : A forward-chaining rule-based inference engine is used to evaluate visa and ETA eligibility. Immigration regulations are encoded as conditional IF THEN rules structured around three key variables: tourist nationality, intended travel duration, and declared purpose of visit. The forward-chaining mechanism evaluates all applicable rules against the tourist's profile attributes and produces a compliance verdict in real time. This approach is selected over machine learning

classification because immigration eligibility rules are deterministic and formally defined by law they do not require
probabilistic inference and demand 100% rule fidelity with no tolerance for prediction error.

Natural Language Processing Named Entity Recognition (NER) : A Transformer-based NER model is used to automatically extract compliance-relevant attributes from free-text tourist inputs and itinerary data. The model identifies and classifies entities including nationality references, travel dates, duration expressions, and purpose-of-visit keywords from unstructured natural language inputs. Pre-trained multilingual transformer models such as BERT or spaCy's transformer pipeline are fine-tuned on a tourism-domain dataset to improve extraction accuracy for Sri Lanka-specific travel contexts.

Knowledge Base Design : Sri Lanka's visa and ETA regulations are encoded into a structured, versioned regulatory knowledge base following the LegalRuleML ontological framework. The knowledge base stores nationality-eligibility mappings, required documentation lists, authorized stay durations, fee structures, and processing timelines in a machine-readable JSON/XML format that can be dynamically queried by the rule-based engine and updated independently when regulations change.

Automated Alert Engine :An event-driven notification engine monitors confirmed travel dates and visa validity periods against system timestamps. When a defined threshold is crossed such as 30 days, 14 days, or 7 days before a compliance deadline the engine triggers automated mobile push notifications to the tourist's device. The alert logic is implemented using a scheduled background service integrated with Firebase Cloud Messaging (FCM) for mobile push delivery.

3.3 Tools and Platforms with Justification

Tool / Platform	Purpose	Justification
React Native	Mobile application development	Cross-platform single codebase for iOS and Android; widely adopted for tourist-facing mobile apps
Node.js + Express.js	Backend server and REST API	Fast, lightweight, non-blocking server framework; ideal for handling real-time compliance validation requests and REST API routing between mobile app and backend services
Python (Flask)	Rule engine and NLP processing	Lightweight framework with extensive AI/NLP library support via spaCy and HuggingFace; used specifically for rule-based inference and NLP model serving
MongoDB	Regulatory knowledge base and tourist compliance records	NoSQL document database ideal for storing flexible, nested regulatory rule structures that vary by nationality; supports dynamic querying and easy rule updates without schema migration
spaCy / HuggingFace Transformers	NLP profile extraction	Industry-standard NLP libraries with pre-trained multilingual

		transformer models for named entity recognition
Firebase Cloud Messaging (FCM)	Push notification delivery	Free, reliable, real-time push notification service for both iOS and Android devices
REST API	Integration with travel planning engine	Standard, platform-agnostic protocol enabling compliance module to connect with the broader R26-IT-112 system
Postman	API testing	Industry-standard tool for REST API functional testing and endpoint validation

Figure 3.3 Tools and Platforms summary table

3.4 Data Collection and Ethical Considerations

Data Collection Strategy:
The system uses a hybrid data collection approach combining primary and secondary data sources.

Secondary Data : Sri Lanka's visa and ETA regulatory data is collected from the official Sri Lanka Department of Immigration and Emigration website and official government gazettes. This data forms the foundation of the regulatory knowledge base and covers eligibility conditions, documentation requirements, authorized stay durations, and fee structures for all applicable nationalities. SLTDA statistical reports are used to validate tourist source market demographics used in testing scenarios.

Primary Data : Simulated tourist profile datasets are constructed to represent realistic inputs from the top 10 tourist source nationalities visiting Sri Lanka, including Indian, Russian, British, German, and Chinese travelers. These synthetic profiles are used to test the NLP extractor and rule-based engine against known compliance verdicts. A small user acceptance testing (UAT) group of 15-20 participants including travel agents, university students with international travel experience, and academic experts will be recruited to evaluate system usability, checklist clarity, and alert effectiveness.

Ethical Considerations:
This research strictly adheres to the Sri Lanka Personal Data Protection Act (No. 9 of 2022) and SLIIT Ethics Review Committee (ERC) guidelines. All primary data collected during UAT is anonymized and no personally identifiable information is retained beyond the testing period. Participants provide informed consent through a formal opt-in mechanism before engaging with the system. No real passport data, visa records, or biometric information is collected or processed at any stage of development or testing. All regulatory data used is sourced exclusively from publicly available official government sources. Formal ERC approval will be sought prior to conducting any user testing activities.

3.5 Validation Strategy and Performance Metrics

The system is validated through three complementary evaluation approaches:

Functional Testing : Each system component is tested independently against a predefined test case library covering all major nationality-visa combinations in the regulatory knowledge base. The rule-

based engine is evaluated against 50 manually verified compliance scenarios covering eligible, ineligible, and edge-case tourist profiles. Pass criteria: minimum 90% compliance verdict accuracy.

NLP Extraction Testing : The NER model is evaluated using precision, recall, and F1-score metrics on a held-out test dataset of 100 simulated tourist profile inputs. Target performance: F1-score ≥ 0.85 for all entity types (nationality, dates, duration, purpose).

User Acceptance Testing (UAT) : 15–20 participants interact with the mobile application through structured test scenarios and complete a standardized usability questionnaire based on the System Usability Scale (SUS). Target: SUS score ≥ 70 (above average usability threshold).

Integration Testing : The compliance module is tested within the broader R26-IT-112 system to verify that itinerary validation correctly blocks legally infeasible trip confirmations and that alert notifications are delivered within defined time thresholds.

Performance Metrics Summary:

Component	**Metric**	**Target**
Rule-Based Engine	Compliance verdict accuracy	≥ 90%
NLP Extractor	F1-score	≥ 0.85
Checklist Generator	UAT usability (SUS score)	≥ 70
Alert Engine	Notification delivery time	≤ 5 seconds
Itinerary Validator	Infeasible itinerary block rate	100%
Overall System	UAT satisfaction rate	≥ 80%

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The AI-Based Immigration and Travel Compliance Management System operates as a dedicated module within the broader Smart Tourism Management System. It processes compliance data inputs and produces verified visa/ETA checklists that are fed back into the main travel planning engine.

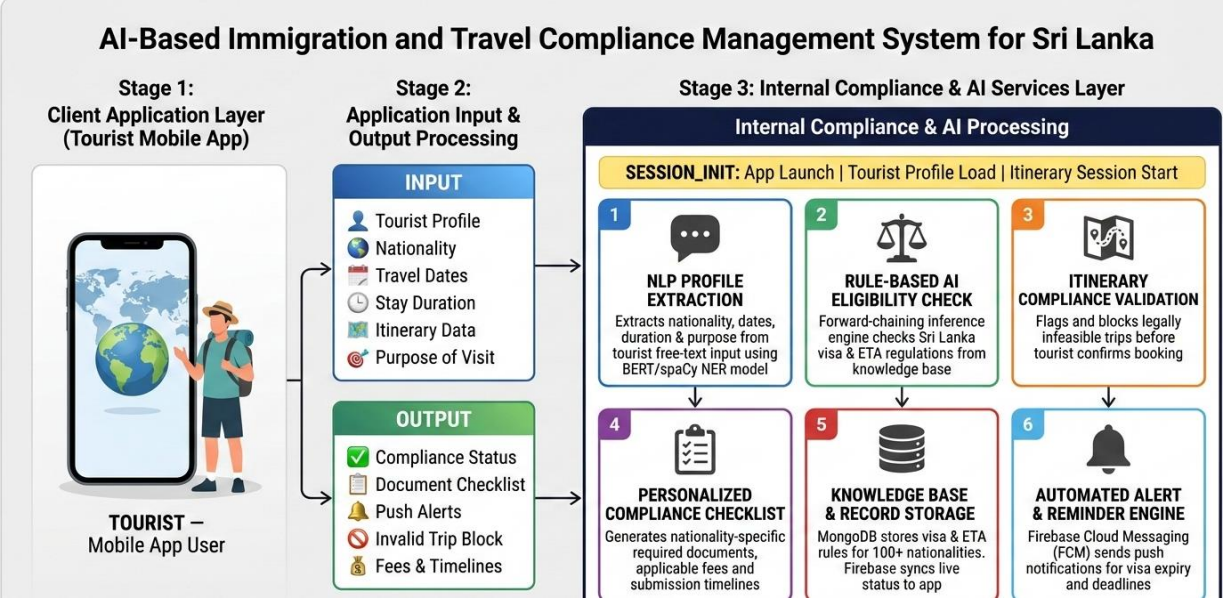


Figure 4.1 High-Level System Architecture AI-Based Immigration and Travel Compliance Management System

4.2 Overview of Proposed Solution and Its Components

The proposed solution is an AI-Based Immigration and Travel Compliance Management System delivered as a mobile application built on React Native, supported by a Node.js and Express.js backend, a Python Flask AI processing layer, and a MongoDB database. The system is designed as a dedicated compliance module within the broader Smart Tourism Management System (Project R26-IT-112), integrating seamlessly with the AI travel planning engine via RESTful APIs. The solution eliminates the current reliance on static government portals and generic third-party visa helpers by providing tourists with real-time, nationality-specific, and itinerary-aware compliance guidance from a single mobile interface.

The system comprises five core components that work together to deliver end-to-end compliance management. The NLP-Based Profile Extractor is responsible for reading free-text tourist input and automatically identifying key travel attributes including nationality, travel dates, intended duration of stay, and purpose of visit. It uses a pre-trained transformer-based Named Entity Recognition model built on BERT or spaCy to eliminate the need for manual structured data entry by the tourist. The Regulatory Knowledge Base is a structured MongoDB database containing Sri Lanka's visa and ETA rules for over one hundred nationalities, encoding eligibility conditions, required documentation, applicable fees, permitted stay durations, and special category restrictions sourced directly from the Department of Immigration and Emigration. The Rule-Based AI Decision Engine is a forward-chaining inference engine implemented in Python Flask that evaluates extracted tourist profile attributes against the Regulatory Knowledge Base using deterministic IF-THEN logic, producing a clear compliance verdict and identifying any violations or eligibility risks. The Compliance Checklist Generator translates the engine's verdict into a personalized, human-readable checklist of required documents, fees, and submission deadlines specific to the tourist's nationality and travel plan. The Automated Alert and Reminder Engine is an event-driven scheduler integrated with Firebase Cloud Messaging that monitors confirmed travel dates and proactively sends push notifications to the tourist's device when visa expiry dates, application deadlines, or documentation submission windows are approaching.

4.2 Individual Component Highlight and Its Parts

Component 1: NLP-Based Profile Extractor. This component is the entry point of the system. It consists of a text pre-processing pipeline that tokenises and cleans tourist input, a Named Entity Recognition layer that identifies and labels entities such as country of origin, travel dates, and visit purpose, and a structured output serialize that converts extracted entities into a JSON profile object consumed by the downstream decision engine. The NER model is fine-tuned on travel and immigration domain vocabulary using the HuggingFace Transformers library and hosted within the Python Flask microservice.

Component 2: Regulatory Knowledge Base. The knowledge base is structured as a MongoDB document store and contains regulatory records for over one hundred nationalities. Each record encodes the visa category applicable to that nationality, the maximum permitted stay duration, the list of mandatory supporting documents, applicable ETA and visa fees, and any special conditions or restrictions. The schema is version-controlled and supports administrative updates without system downtime, ensuring that rule changes issued by the Department of Immigration and Emigration can be applied promptly and safely.

Component 3: Rule-Based AI Decision Engine. The decision engine implements a forward-chaining inference mechanism. It receives the structured tourist profile from the NLP extractor, queries the knowledge base for the matching nationality record, and sequentially evaluates a ruleset covering visa eligibility, duration compliance, document completeness, and fee obligations. Each rule produces a pass, fail, or warning outcome, and the aggregated result forms a full compliance verdict. The deterministic nature of forward-chaining ensures 100% rule fidelity and full auditability of every decision made by the engine.

Component 4: Compliance Checklist Generator. This component receives the compliance verdict and generates a plain-language, nationality-specific checklist presented to the tourist within the mobile application. The checklist itemizes required documents, applicable ETA or visa fees, submission deadlines, and any special instructions relevant to the tourist's specific travel plan. The output is stored in MongoDB as a compliance record linked to the tourist's profile, enabling historical reference and audit support.

Component 5: Automated Alert and Reminder Engine. The alert engine is an event-driven scheduler that monitors confirmed itinerary dates stored in MongoDB and computes the time remaining until critical compliance deadlines, including visa application submission windows, ETA expiry dates, and entry validity periods. When a threshold is reached, the engine triggers a push notification via Firebase Cloud Messaging directed to the tourist's registered mobile device. Alert thresholds are configurable at 30 days, 7 days, and 24 hours before each deadline, ensuring tourists receive timely and escalating reminders.

4.3 Component Interactions

The five components interact through a structured request-response pipeline coordinated by the Node.js and Express.js API gateway. The mobile application communicates exclusively with the API gateway via HTTPS REST calls, and the gateway routes requests to the appropriate backend service. When a

tourist submits their profile, the API gateway forwards the raw input to the Python Flask microservice hosting the NLP Profile Extractor. The extractor returns a structured JSON profile to the gateway, which immediately passes it to the Rule-Based AI Decision Engine within the same Flask service. The engine queries the MongoDB Regulatory Knowledge Base, evaluates the applicable rule set, and returns a compliance verdict object back to the gateway. The gateway then invokes the Compliance Checklist Generator, which constructs the personalised checklist from the verdict and persists the record to MongoDB. The final checklist is returned to the mobile application for display. Concurrently, the Itinerary Validation Gateway a dedicated API endpoint intercepts any itinerary confirmation request from the main travel planning system and checks the stored compliance verdict before permitting booking confirmation. If a compliance failure is detected, the gateway returns a block response that prevents the invalid itinerary from being confirmed. Following successful itinerary confirmation, the Automated Alert Engine is activated and begins scheduled monitoring of the tourist's compliance deadlines, dispatching Firebase Cloud Messaging push notifications at configured intervals until all compliance obligations are fulfilled.

4.4 Data Flow Explanation

The data flow through the system follows a six-stage sequence. In Stage 1, the tourist enters their travel information as free text into the React Native mobile application, including nationality, intended travel dates, duration of stay, and purpose of visit. This raw input is transmitted as an HTTPS POST request to the Node.js and Express.js API gateway in Stage 2. In Stage 3, the gateway forwards the input to the Python Flask NLP service, where the Named Entity Recognition model processes and extracts structured travel attributes, returning a typed JSON profile object. Stage 4 involves the Rule-Based Decision Engine querying the MongoDB Regulatory Knowledge Base with the extracted nationality and travel parameters, evaluating the applicable visa and ETA ruleset, and generating a compliance verdict that is returned to the gateway. In Stage 5, the Compliance Checklist Generator constructs a personalised checklist from the verdict, stores it as a compliance record in MongoDB, and the complete result is returned to the mobile application via the gateway for display to the tourist. Firebase Real-Time Sync ensures that any update to the compliance status is immediately reflected on the tourist's device without requiring a manual refresh. In Stage 6, once the itinerary is confirmed, the Automated Alert Engine reads the compliance record from MongoDB, calculates all relevant deadline windows, and registers scheduled notification tasks that trigger Firebase Cloud Messaging push alerts to the tourist's device at 30-day, 7-day, and 24-hour intervals before each critical compliance deadline.

4.5 Scalability and Security Considerations

The system is designed with scalability in mind to accommodate Sri Lanka's seasonal tourism peaks, during which compliance request volumes may spike significantly. The Node.js API gateway is stateless and can be horizontally scaled across multiple instances behind a load balancer without service interruption. The Python Flask AI processing service is independently deployable and scalable, allowing the NLP and rule engine workload to be distributed across additional instances during high-demand periods. MongoDB's document-oriented architecture supports horizontal sharding, enabling the Regulatory Knowledge Base and compliance records to scale across distributed nodes as tourist user volumes grow. Firebase Real-Time Database provides inherently scalable synchronization for live compliance status updates across all connected mobile clients.

From a security perspective, all data transmitted between the mobile application and the API gateway is encrypted in transit using TLS 1.3. All compliance records and tourist profile data stored in MongoDB are encrypted at rest using AES-256 encryption. Authentication is enforced through JSON Web Token-

based middleware on the API gateway, ensuring that only verified users can access compliance endpoints. Role-based access control restricts regulatory knowledge base update operations exclusively to authorized system administrators, preventing unauthorized modification of immigration rule data. The knowledge base is version-controlled, enabling safe rollback in the event that an incorrect regulatory update is applied. No real passport numbers, biometric data, or personally identifiable documentation is stored by the system; only the extracted structured travel attributes required for compliance evaluation are persisted, in full compliance with the Sri Lanka Personal Data Protection Act No. 9 of 2022.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The AI-Based Immigration and Travel Compliance Management System serves five distinct stakeholder groups, each with different needs and levels of interaction with the system.

Foreign Tourists are the primary end users of the system. They interact directly with the mobile application to input travel details, receive compliance verdicts, view personalized checklists, and receive push notification alerts. Their core need is accurate, real-time, and easy-to-understand immigration guidance that is fully integrated into their travel planning experience without requiring them to consult external government portals or visa agencies.

Tour Operators are secondary users who benefit from the system through the Itinerary Validation Gateway integrated into the travel planning system. They require assurance that all itineraries sold to international clients are legally feasible and immigration-compliant before booking confirmation, reducing their exposure to legal liability and reputation damage caused by client visa violations.

The Sri Lanka Department of Immigration and Emigration is an indirect stakeholder that serves as the primary data source for the Regulatory Knowledge Base. The department benefits from the system through reduced tourist immigration violations, improved voluntary compliance, and decreased pressure on immigration enforcement resources at ports of entry.

The Sri Lanka Tourism Development Authority (SLTDA) is a policy-level indirect stakeholder. The SLTDA benefits from the system through improved compliance statistics that support Sri Lanka's international tourism reputation, reduced visa violation incident rates, and stronger alignment with the national goal of reaching three million tourist arrivals as reported in the SLTDA Year in Review 2024 [3].

System Administrators are internal technical users responsible for maintaining the Regulatory Knowledge Base, applying regulatory updates issued by the Department of Immigration and Emigration, monitoring system health, and managing role-based access controls. They interact with the system through a secured administrative interface that supports version-controlled knowledge base updates without requiring system downtime.

5.2 Functional Requirements

The following functional requirements define the core capabilities the system must deliver to satisfy stakeholder needs. Each requirement is stated using the standard "shall" convention to indicate a mandatory system obligation.

FR-01: Profile Extraction. The system shall automatically extract tourist nationality, travel dates, intended duration of stay, and purpose of visit from free-text user input using a transformer-based NLP Named Entity Recognition model, requiring zero manual structured data entry from the tourist.

FR-02: Visa and ETA Eligibility Evaluation. The system shall evaluate visa and ETA eligibility in real time by applying a forward-chaining rule-based AI engine against the Regulatory Knowledge Base, covering nationality, travel duration, visit purpose, and applicable visa category for a minimum of one hundred nationalities.

FR-03: Compliance Checklist Generation. The system shall generate a personalized, nationality-specific compliance checklist for every tourist, clearly listing all required documents, applicable ETA or visa fees, submission timelines, and any special conditions relevant to their travel plan, presented in plain language within the mobile application interface.

FR-04: Itinerary Compliance Validation. The system shall intercept all itinerary confirmation requests from the travel planning engine via a dedicated REST API gateway and block confirmation of any itinerary that is determined to be legally infeasible under the tourist's visa or ETA category, returning a descriptive violation reason to the calling system.

FR-05: Automated Deadline Alerts. The system shall send automated push notifications to the tourist's registered mobile device at 30-day, 7-day, and 24-hour intervals before each critical compliance deadline, including visa application submission windows, ETA expiry dates, and entry validity periods, using Firebase Cloud Messaging.

FR-06: Regulatory Knowledge Base Management. The system shall maintain a centralized, versioned MongoDB database of Sri Lanka visa and ETA regulations covering a minimum of one hundred nationalities, supporting administrative updates to regulatory rules without requiring system downtime or redeployment.

FR-07: Compliance Record Storage. The system shall persist each tourist's compliance verdict, generated checklist, and alert history as a structured compliance record in MongoDB, enabling historical reference, audit support, and repeat session retrieval without requiring the tourist to re-enter their profile information.

FR-08: Real-Time Status Synchronization. The system shall synchronize the tourist's live compliance status to their mobile device in real time via Firebase Real-Time Database, ensuring that any change to compliance state is immediately reflected in the application without requiring a manual page refresh.

FR-09:System Integration. The system shall expose a set of documented RESTful API endpoints that

allow the main Smart Tourism Management System to query compliance status, submit itinerary validation requests, and receive compliance verdicts programmatically, enabling seamless cross-component integration within the project group platform.

5.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints the system must satisfy independently of specific functional behaviors. Six categories are identified below.

NFR-01: Performance. The end-to-end compliance validation pipeline, from receipt of the tourist's profile input at the API gateway to delivery of the compliance verdict and checklist to the mobile application, shall complete within two seconds under normal operating load. The NLP profile extraction step alone shall complete within one second. Push notification delivery via Firebase Cloud Messaging shall occur within five seconds of the scheduled trigger time. These thresholds are informed by established mobile application usability standards and the performance benchmarks reported by Vadisetty et al. (2026) [5] for AI compliance monitoring systems.

NFR-02: Accuracy. The rule-based AI decision engine shall achieve a minimum compliance determination accuracy of ninety percent across the full set of simulated test scenarios covering the ten highest-volume tourist nationalities. The NLP profile extractor shall achieve a minimum F1-score of 0.85 on the Named Entity Recognition task for the defined entity types. The itinerary validation gateway shall achieve a one hundred percent block rate for itineraries confirmed as legally infeasible by the rule engine, with zero false negatives permitted.

NFR-03: Security. All data transmitted between the mobile application and the API gateway shall be encrypted in transit using TLS 1.3. All tourist compliance records and profile data stored in MongoDB shall be encrypted at rest using AES-256. Authentication to all API endpoints shall be enforced via JSON Web Token middleware. Role-based access control shall restrict knowledge base update operations to authorized administrators only. No passport numbers, biometric data, or raw identity documents shall be stored at any point. The system shall comply fully with the Sri Lanka Personal Data Protection Act No. 9 of 2022.

NFR-04: Usability. All compliance checklists and alert messages shall be presented in plain, non-technical language that is comprehensible to international tourists without legal or immigration expertise. The mobile application interface shall achieve a minimum System Usability Scale (SUS) score of seventy, as established by the TAM usability benchmarks reported in Davis (1989) [6], in User Acceptance Testing conducted with a sample group of fifteen to twenty international tourist participants. The application shall support English as the primary language, with provision for Sinhala, Tamil, Chinese, and Hindi checklist display in later iterations.

NFR-05: Availability. The system shall maintain a minimum uptime of 99.5% during Sri Lanka's peak tourist seasons, specifically the December-to-March and July-to-August periods identified in the SLTDA Annual Statistical Report 2023 [3]. The Node.js API gateway and Python Flask AI service shall be independently deployable and horizontally scalable to accommodate seasonal demand spikes without service interruption.

NFR-06: Maintainability. The Regulatory Knowledge Base shall be updatable by authorized administrators through a versioned update mechanism that applies regulatory changes without requiring system redeployment or downtime. All knowledge base versions shall be retained to support safe rollback in the event that an incorrect rule update is applied. The system codebase shall be modular and follow RESTful microservice architecture principles, enabling individual components to be updated, tested, and redeployed independently of one another.

6. COMMERCILIAZION PLAN

6.1 Target Market

The primary target market consists of foreign tourists planning visits to Sri Lanka, particularly the 2.5 million annual arrivals from India, China, and Europe. Secondary markets include Sri Lankan tour operators, travel agencies, and online travel platforms (OTAs) serving inbound tourism. The system can additionally be licensed to government tourism boards seeking to modernize their digital compliance infrastructure.

6.2 Value Proposition

- Eliminates the most common cause of tourist immigration violations through proactive, embedded compliance checking.
- Reduces visa rejection and overstay risks, improving Sri Lanka's international tourism reputation.
- Saves tourists an average of 3-5 hours of independent immigration research per trip.
- Provides tour operators with compliance assurance that reduces legal liability.

6.3 Revenue Model

- B2B SaaS Licensing: Monthly subscription fee charged to tour operators and OTAs for API access to the compliance engine (estimated LKR 15,000–50,000/month per operator).
- B2G Partnership: Licensing agreement with SLTDA or the Department of Immigration for official integration into national tourism platforms.
- Freemium Consumer Model: Free basic compliance check for individual tourists; premium personalized checklist and real-time alerts available for a nominal fee (USD 2-5 per trip).

6.4 Competitive Advantage

Unlike generic visa helper platforms, this system is the only solution that integrates immigration compliance directly within the travel planning workflow, preventing legally invalid itineraries from being confirmed. Its deep integration with Sri Lanka-specific regulatory data and the broader Smart Tourism ecosystem creates a defensible competitive moat. Intellectual property protection will be sought through copyright registration of the regulatory ontology schema and the rule engine architecture.

7. BUDGET AND JUSTIFICATION

Component	Amount (Rs.)
Internet / Cloud Hosting	10,000.00
Software Tools	8,700.00
Database and Server Cost	1,200.00
Hardware (Android Device)	26,500.00
User Acceptance Testing	10,000.00
Printing and Stationery	4,500.00
Other / Contingency	2,325.00
Total	**63,225.00**

Figure 7.1 Budget summary table

The total estimated budget for the AI-Based Immigration and Travel Compliance Management System is LKR 63,225. Internet and cloud hosting covers MongoDB Atlas, Firebase, and Render.com services used during development and testing. Software tools include a Google Colab Pro subscription for GPU-accelerated NLP model training. Database and server costs relate to cloud storage for training datasets and model files. Hardware costs cover an Android device required for real-device mobile testing and Firebase FCM validation. User Acceptance Testing costs cover participant incentives for fifteen to twenty international tourist participants. Printing and stationery cover the project proposal, progress reports, and UAT consent forms. The other contingency allocation provides a five-percent reserve for any unforeseen minor expenses.

8. WORK BREAKDOWN STRUCTURE (WBS)

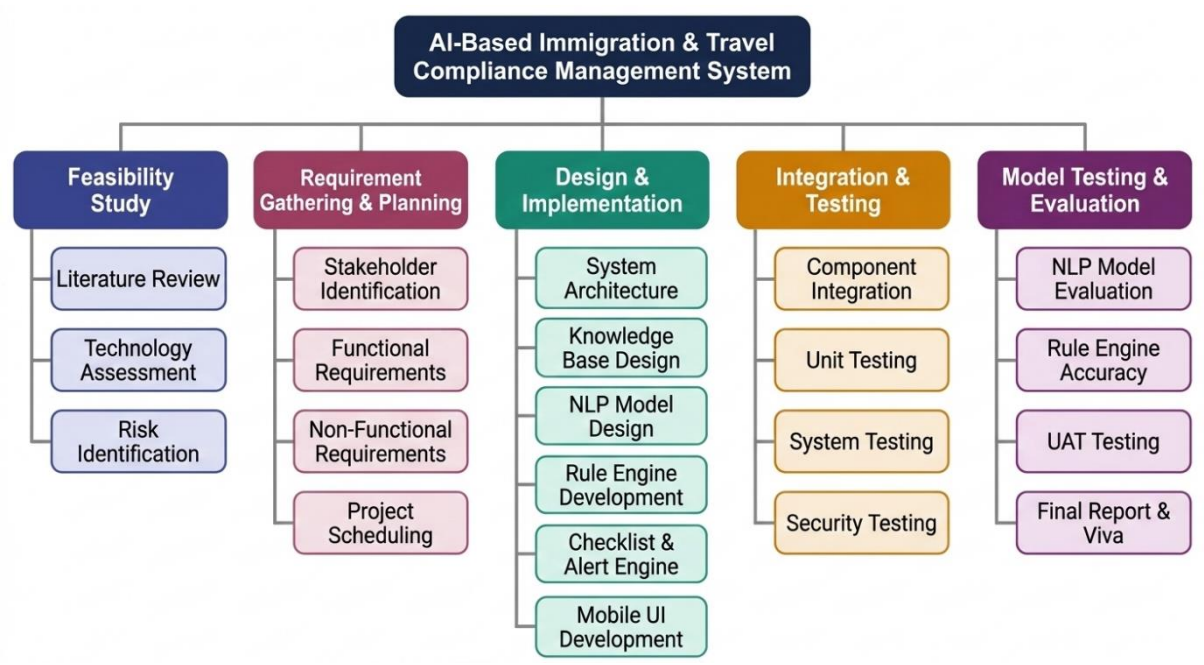


Figure 8.1 WBS Chart

9. GRANT CHART

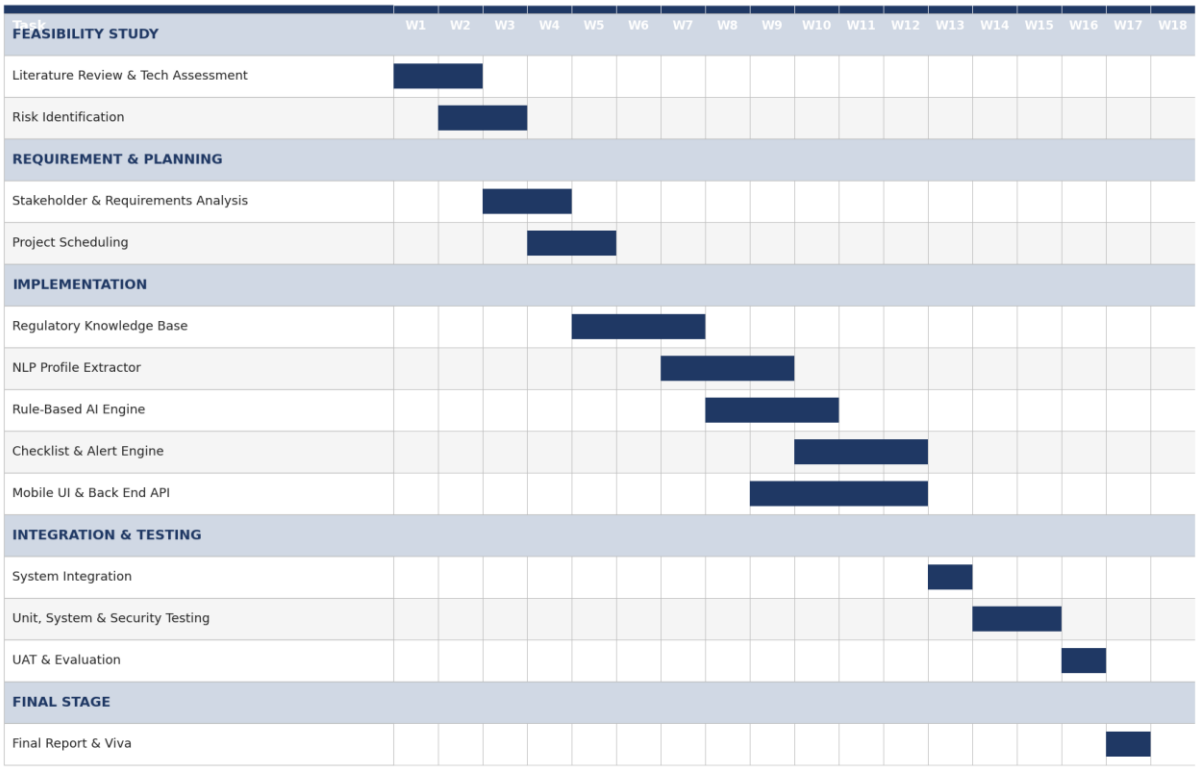


Figure 9.1 Grant Chart

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11. APPENDICES

Risk	Likelihood	Impact	Mitigation
Outdated regulatory data in knowledge base	Medium	High	Implement versioned updates with admin alerts when regulations change
NLP misidentification of nationality	Low	High	Implement fallback manual nationality selection and confidence threshold checks
API integration failures with planning engine	Low	Medium	Design fault-tolerant API with retry logic and graceful degradation
Data privacy breach of tourist profiles	Low	High	Enforce end-to-end encryption and minimal data retention policies
Scope creep due to complex regulatory variations	Medium	Medium	Adopt agile sprints with fixed scope per iteration; defer edge cases

Appendix B: Ethical Compliance Checklist

- No real personal data collected during development phase
- Simulated datasets used for all testing and validation
- System complies with Sri Lanka Personal Data Protection Act (No. 9 of 2022)
- Ethics Review Committee (ERC) approval will be obtained prior to user testing
- Informed consent mechanism built into user facing interface
- Data anonymization applied to all stored compliance records
- High level encryption implemented for all data in transit and at rest

Appendix C: Preliminary UI Mockup Descriptions

The following describes the four primary mobile application screens planned for the system. Full visual mockups will be developed during the Design and Implementation phase using Figma and included in the final dissertation.

Screen 1 : Profile Input Screen
A free-text input field where the tourist enters their travel details in natural language (e.g. nationality, travel dates, purpose of visit). The NLP Profile Extractor processes this input automatically. A simple form fallback is available for manual entry.

Screen 2 : Compliance Status Screen
Displays the compliance verdict (Eligible / Not Eligible / Conditional) with a clear color coded status indicator. Shows the applicable visa or ETA category, permitted stay duration, and any special conditions identified by the Rule-Based AI Decision Engine.

Screen 3 : Compliance Checklist Screen
A scrollable list of all required documents, applicable fees, and submission deadlines generated specifically for the tourist's nationality and travel plan. Each checklist item can be marked as completed. Items are presented in plain language with no legal terminology.

Screen 4: Alerts and Reminders Screen
Displays a chronological list of all active alerts and upcoming compliance deadlines. Shows 30 day, 7 days, and 24-hour reminders for visa expiry, ETA submission windows, and missing document warnings delivered via Firebase Cloud Messaging push notifications.